Rankine Cycle Analysis

EGR 218: Engineering Thermodynamics

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May 5, 2026

Abstract

The Rankine cycle is a common cycle used in power generation. This report analyzes and presents a Rankine cycle power plant using Engineering Equation solver (EES). Several real world examples of the Rankine cycle in use are examined. The study examines an ideal Rankine cycle using a boiler pressure of 8 MPa, condenser pressure of 10 kPa, and a turbine inlet tempreature of 480°C. The results of the study can be quantified by the net work, w_{net}, of the cycle and the thermal efficiency, η_{th}. The net work was 1230 kJ/kg while the thermal efficiency was 39.1%. A parametric study was performed in EES to study the effect of boiler pressure, condenser pressure, and turbine inlet temperature on thermal efficiency, η_{th}, and work output, w_t. The figures generated by this study suggest that the work done by the turbine and the thermal efficiency are effected positively, in a logarithmic manner, by increasing pressure exiting the boiler. The thermal efficiency and work output were also positively affected by an increase in the temperature exiting the boiler, but in a linear manner. Increasing the pressure of the medium exiting the condenser decreased the work output and the thermal efficiency of the system.

Contents

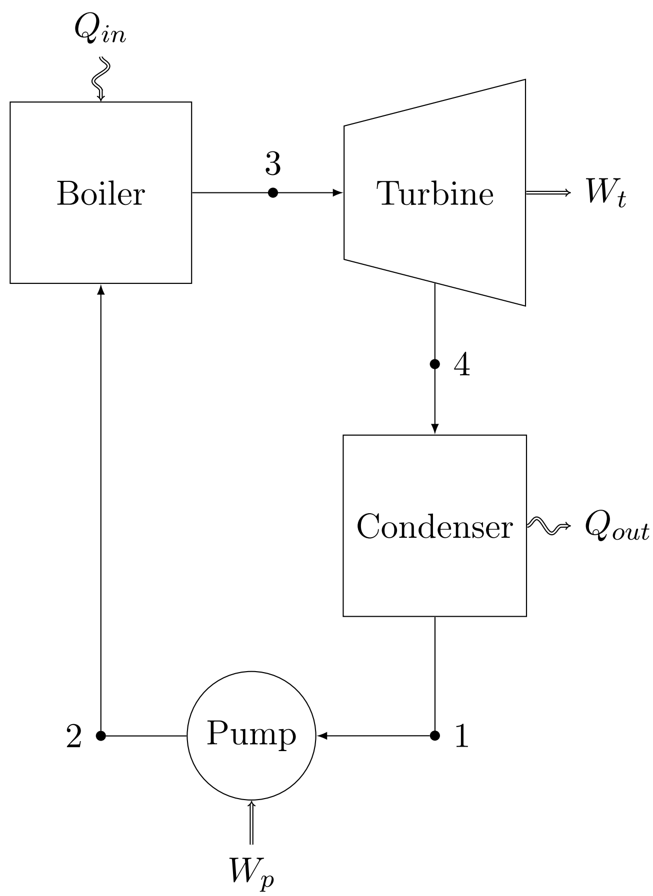

Figure 1: Schematic showing a Rankine cycle.

1 Introduction

1.1 Theoretical Background

The Rankine cycle consists of a boiler, a turbine, a condenser, and a pump, in a loop. An example of a Rankine cycle setup is shown in Figure 1. The Rankine cycle is characterized by its phase changes and work output [1].

The Rankine cycle consists of two state changes, one in the boiler, and one in the condenser. In the boiler the liquid is converted to a saturated vapor and the saturated vapor in the condenser is converted back into a saturated liquid. Heat is added in the boiler, the turbine outputs work, the condenser outputs heat, and work is done on the system by the pump. The pressure is constant from the output of the pump through the boiler to the input of the turbine. The turbine decreases the pressure, this decreased pressure is constant from the output of the turbine through the condenser to the input of the pump. The pump increases the pressure to restart the cycle. The liquid is heated in the boiler and then is cooled through the turbine. The temperature stays constant through the condenser. The temperature rises from the output of the pump to the boiler, due to increasing pressure and work done on the system by the pump. The cycle is isentropic through the pump and through the turbine. These relationships are summarized in Table 1.

The difference between an ideal and actual Rankine cycle is efficiency and reversibility. In an ideal Rankine cycle we can make the assumptions in Table 1, whereas in an actual Rankine cycle all processes are polytropic and you will have heat and pressure loss at points where the ideal cycle assumes you would not. Friction and heat loss are two major places where efficiency is lost, fluid friction can decrease pressure [2].

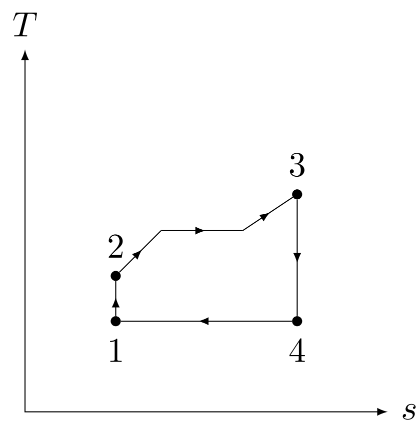

Figure 2: Ideal Rankine Cycle T-s Diagram

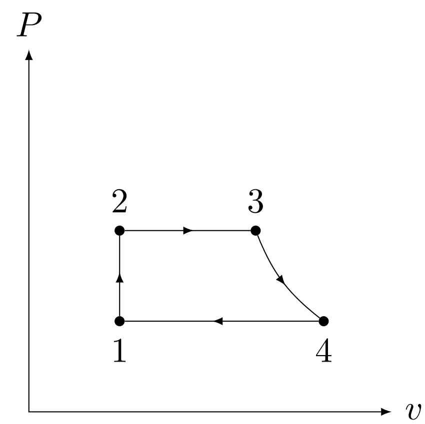

Figure 3: Ideal Rankine Cycle P-v Diagram

The cycle, based on the description given above and the controlled variables listed in Table 1, is governed by the following equations, which are enumerated again in the Problem Setup: $P_1 = P_4$, $P_2 = P_3$, $s_1 = s_2$, $s_3 = s_4$, $T_1 = T_4$, $x_1 = 0.0$, $x_3 = 1.0$, $w_t = h_3 - h_4$, $w_p = h_2 - h_1$, $q_{\text{in}} = h_3 - h_2$, $q_{\text{out}} = h_4 - h_1$, $w_{\text{net}} = q_{in} - q_{out}$, and $\eta_{\text{th}} = \frac{w_{\text{net}}}{q_{\text{in}}}$.

Points	Component	Variables
1-2	Pump	Isochoric, Isentropic.
2-3	Boiler	Isobaric.
3-4	Turbine	Isentropic.
4-1	Condenser	Isochoric, Isothermal.

Table 1: Summary of components and their locations.

1.2 Real World Examples

Figure 4: An aerial view of the Laguna Verde power plant in Veracruz, Mexico. [3]

The Laguna Verde Nuclear Power Plant is a boiling water reactor in Veracruz, Mexico. The reactor was designed, unfortunately, by General Electric, not Westinghouse. The power plant provides for roughly 4.8% of Mexico's electricy production [4] and is the only nuclear power plant in the nation. There are reactors at the site. The reactors follow the Rankine cycle. Water is pumped into the reactor vessel where it is heated by the reactor itself, this is our boiler, the increased pressure steam then turns turbines before it is cooled back down, condensed, and reintroduced to the cycle. A NRC report [5] says the nuclear reactor's system "includes a direct cycle, forced circulation, boiling water reactor that produces steam for direct use in the steam turbine." This description highlights several components of our Rankine Cycle: forced circulation describes the work input done by the pump, boiling water reactor describes the boiler which increases pressure and temperature, and the turbine which produces the work output. Another key factor in this description is the reactor as being direct cycle. A direct cycle reactor passes water directly through the reactor, meaning the water is iraddiated. The water being irradiated means that the cycle must be closed; this gives us our final component, the condenser which converts the steam back into water. The principal aspect of the Rankine cycle is a phase change, and secondarily work output. The reactor's cycle fills these two aspects, in the reactor boiling the water and the turbine extracting work.

Figure 5: A picture of the Westinghouse 501F gas turbine which is very similar to the kind used in the Sayreville Energy Center. [6]

Another example of the Rankine cycle being used in power plants is the Sayreville Energy Center in Sayreville, New Jersey. The energy center was built by Westinghouse and Siemens Energy but is now operated by NextEra Energy Resources [7]. The site uses a Combined Cycle Gas Turbine. The primary cycle of this type of power plant is the Brayton cycle, but the Rankine cycle is used for bottoming, to increase the efficiency of the plant. The first cycle burns the gas and this turns turbines which produce electricity. In the second cycle, the bottoming cycle, the exhaust is used to boil water which then is used to turn turbines and is recycled. This second cycle is a Rankine cycle. This usage is critical to increasing the efficiency of power plants and can increase the efficiency by around 50% [8].

2 Problem Setup

The given variables for our power plant analysis are the following:

Boiler pressure, $P_3 = 8$ MPa, condenser pressure, $P_1 = 10$ kPa, turbine inlet temperature, $T_3 = 480°$C.

We can begin filling out the rest of the state variables by realizing that $P_2 = P_3$, $P_1 = P_4$, these are shown in Figures 2 and 3. As stated in the introduction, the cycle is isentropic through the pump, Point 1 to Point 2, and through the turbine, Point 3 and Point 4. This shows that $s_1 = s_2$ and $s_3 = s_4$. The boiler converts the saturated liquid into a saturated vapor, $x_3 = 1.0$, after the condenser the medium is a saturated liquid, $x_1 = 0.0$.

The rest of the state variables were calculated in EES, alternatively these values could be looked up in state tables. Table 3 displays the results.

Point	P, (kPa)	T, (°C)	x	s, (kJ/(°K·kg))	h, (kJ/kg)	v, (m^3/kg)
1	10.0	45.8	0.00	0.65	192	0.00101
2	8,000	46.1	-100[1]	0.65	200	0.00101
3	8,000	480	1.00	6.67	3350	0.0404
4	10.0	45.8	0.80	6.67	2110	11.76

Table 2: Summary of state variables.

3 Energy Analysis

Variable	value
w_t	1240 kJ/kg
w_p	8.06 kJ/kg
q_{in}	3150 kJ/kg
q_{out}	1920 kJ/kg
w_{net}	1230 kJ/kg
η_{th}	39.1%

Table 3: Summary of energy analysis.

The equations, taken from a SFU lecture [9], used to analyze the power plant Rankine cycle are listed below, the actual calculation was done in EES:

To calculate the turbine work output, w_t:

$$w_t = h_3 - h_4$$

To calculate the pump work input, w_p:

$$w_p = h_2 - h_1$$

To calculate the heat input to the boiler, q_{in}:

$$q_{\text{in}} = h_3 - h_2$$

To calculate the heat rejected in the condenser, q_{out}:

$$q_{\text{out}} = h_4 - h_1$$

[1] EES outputs a quality value of -100 for compressed liquids.

To calculate the net-work output from the cycle, w_{net}:

$$w_{\text{net}} = q_{in} - q_{out}$$

To calculate the efficiency of the cycle, η_{th}:

$$\eta_{\text{th}} = \frac{w_{\text{net}}}{q_{\text{in}}}$$

3.1 Diagrams

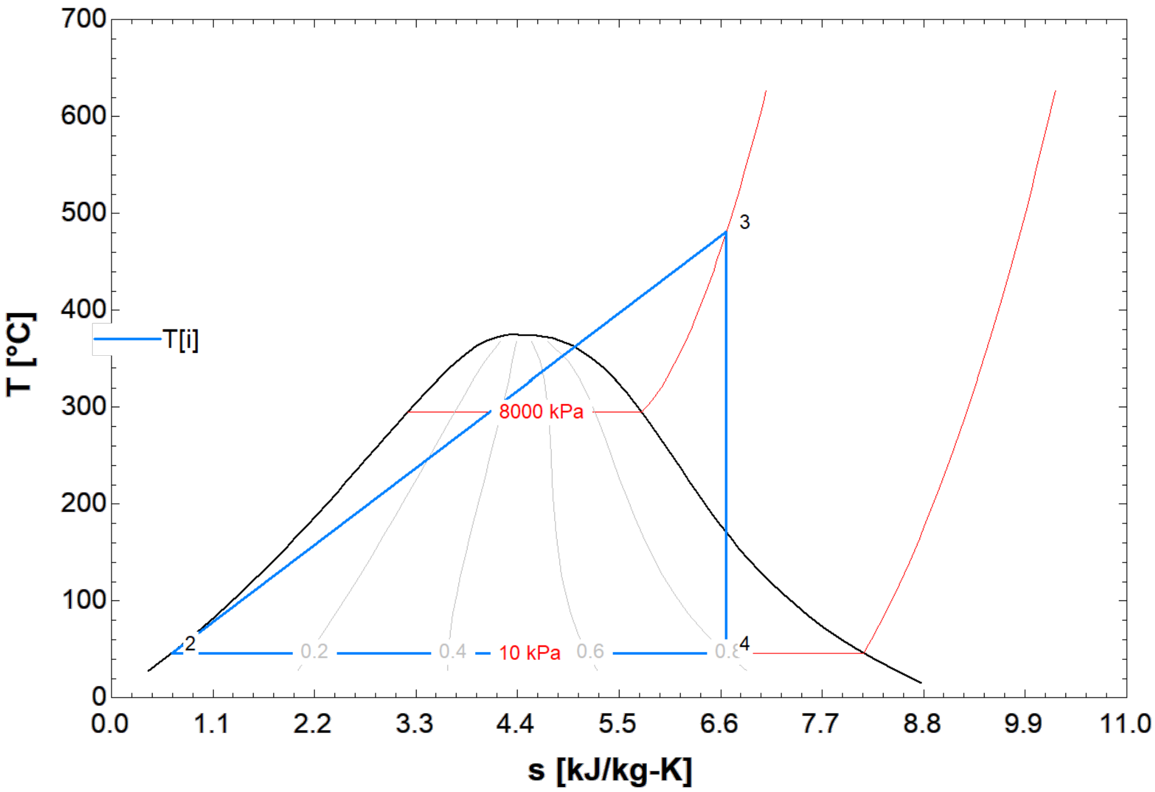

Figure 6: T-s diagram of the Rankine cycle using water. Point 1 is hidden by Point 2 as they are, relative to the scale, close together. The blue lines approximately show the cycle path. The red line at $P = 8000$ kPa is a more accurate representation of the path, as Figure 3 shows us that pressure is constant from Point 2 to Point 3, and Figure 2 demonstrates that the shape of the red line is what we should expect. The black line is the saturation bell.

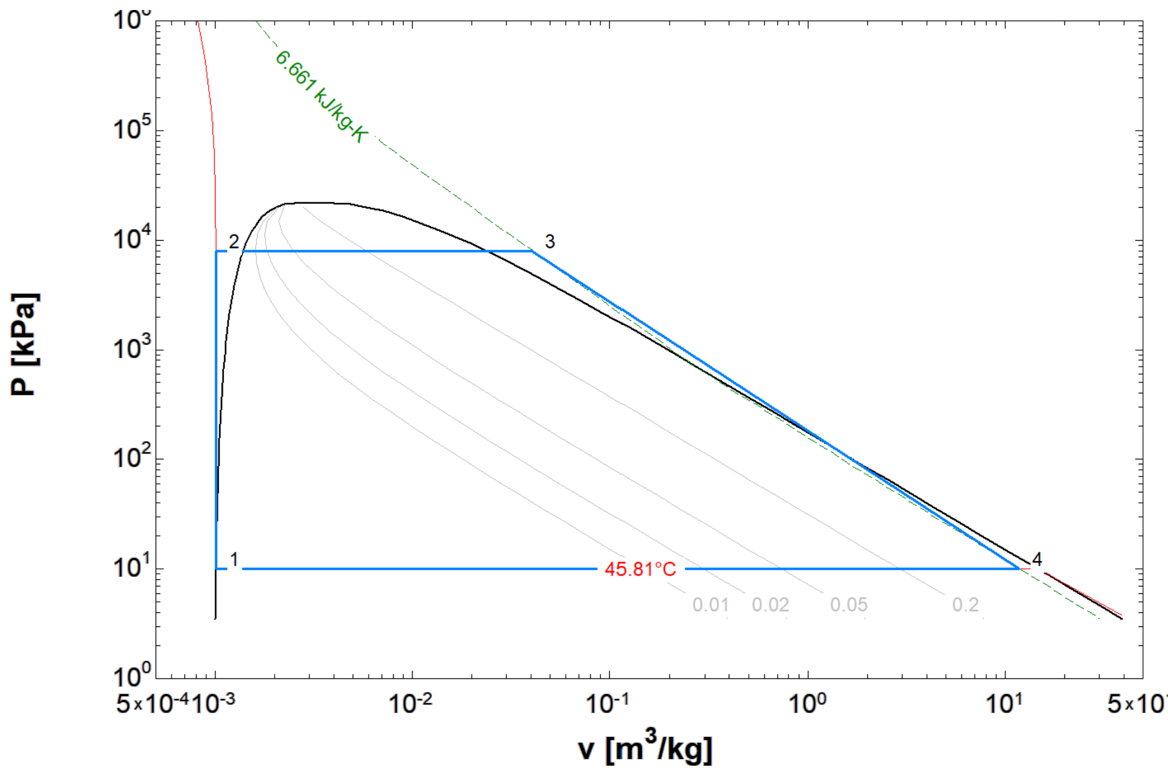

Figure 7: P-v diagram of the Rankine cycle using water. The blue lines approximately show the cycle path. The cycle from Point 3 to Point 4 is better shown by the entropy line, which is shown dotted in green, because entropy at these points is constant. The black line is the saturation bell.

Figure 7 demonstrates the principal factor of the Rankine cycle, the phase changes. In Figure 7 the cycle path starts at Point 1 with a quality of 0.0, a saturated liquid, it climbs outside, to the left of, the saturation bell at Point 2 where it becomes a compressed liquid, compressed by the pump. From Point 2 to Point 3 the path enters back under the saturation bell and comes outside above and to the right of saturation bell; the boiler converts the compressed liquid into a saturated vapor with quality 1.0. From Point 3 to Point 4 it enters back under the saturation bell, with a quality of 0.80.

4 Parametric Study

4.1 Parametric Analysis

A Parametric study is a "what if" analysis utilized in thermodynamics to quantify the affects of progressively incrementing a variable in a given system. Instead of simply analyzing an engine or a powerplant through a P-v or a T-s diagram, as a system with constant and changing variables throughout cycle states; a parametric study systematically isolates variables, measuring the sensitivity of the system. Thus, it can be accurately stated that a parametric study is a premier method for an engineer to optimize a thermodynamic system.

In the specific context of the Rankine cycle, parametric studies can be conducted by progressively modifying varibales at the respective inlets and outlets of the pump, the boiler, the turbine, and the condenser. The parametric studies displayed below in Figures 8-13, analyze the effects of progessively changing pressure leaving the boiler, temperature leaving the boiler, and pressure leaving the condenser. The values of these respective properties at these critical points in the Rankine cycle, notably affect thermal efficiency and work out (turbine work, Figure 1) of the system. Thermal efficiency and net work output (turbine work) are key quantities in a Rankine cycle. In resource management, thermal efficiency defines the ratio of net benifit to total cost. While turbine work serves as the "money maker" quantity; as its output serves as raw mechanical energy.

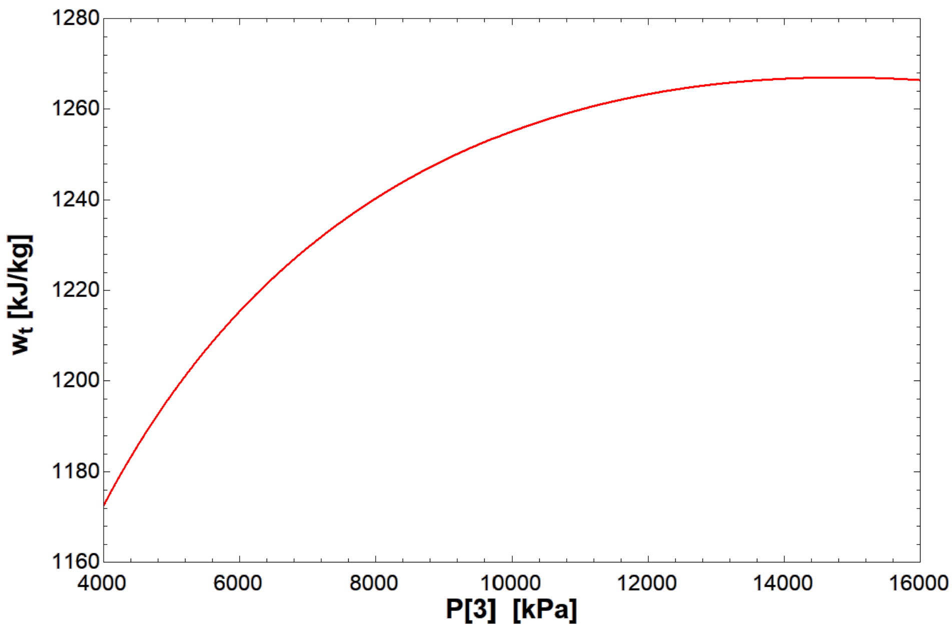

Figure 8: Work done by the turbine as affected by a variable pressure exiting the boiler.

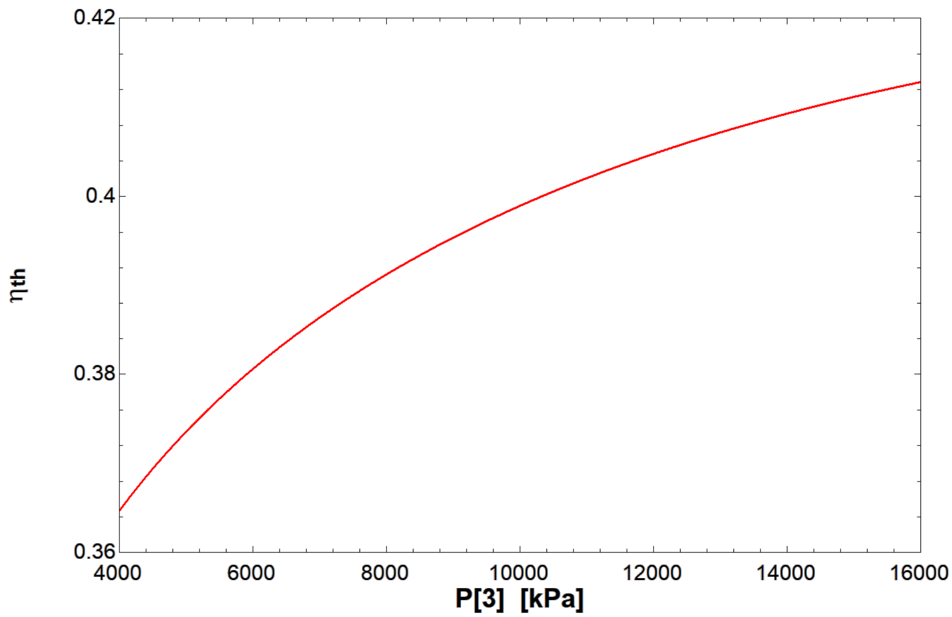

Figure 9: Thermal efficiency as affected by a variable pressure leaving the boiler

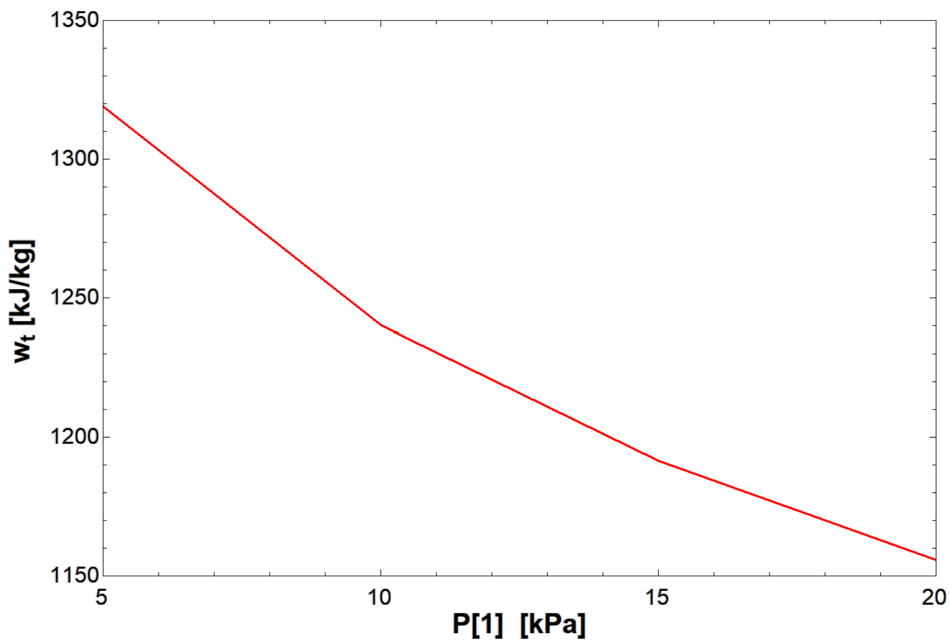

Figure 10: Work done by the turbine as affected by a variable pressure leaving the condenser

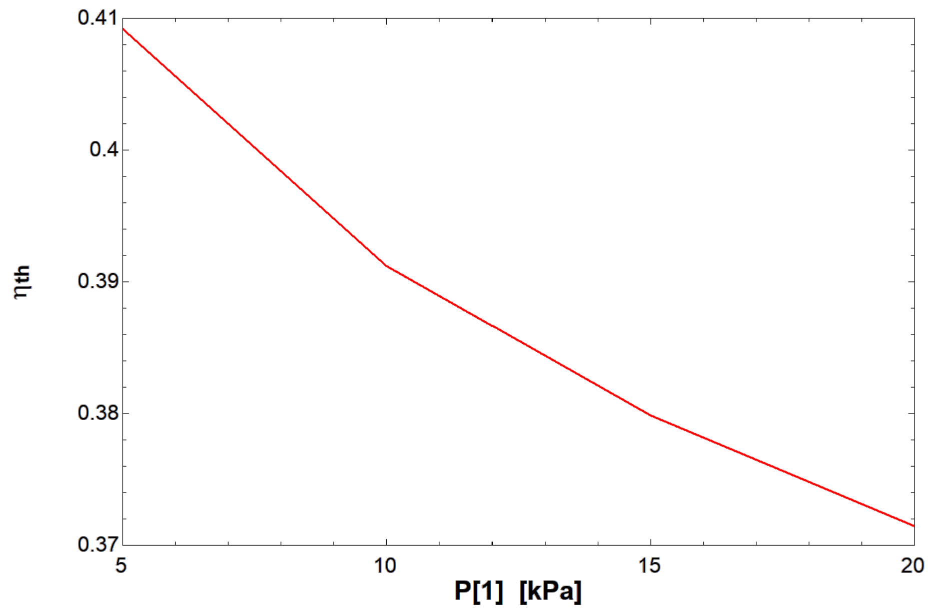

Figure 11: Thermal efficiency as affected by a variable pressure leaving the condenser

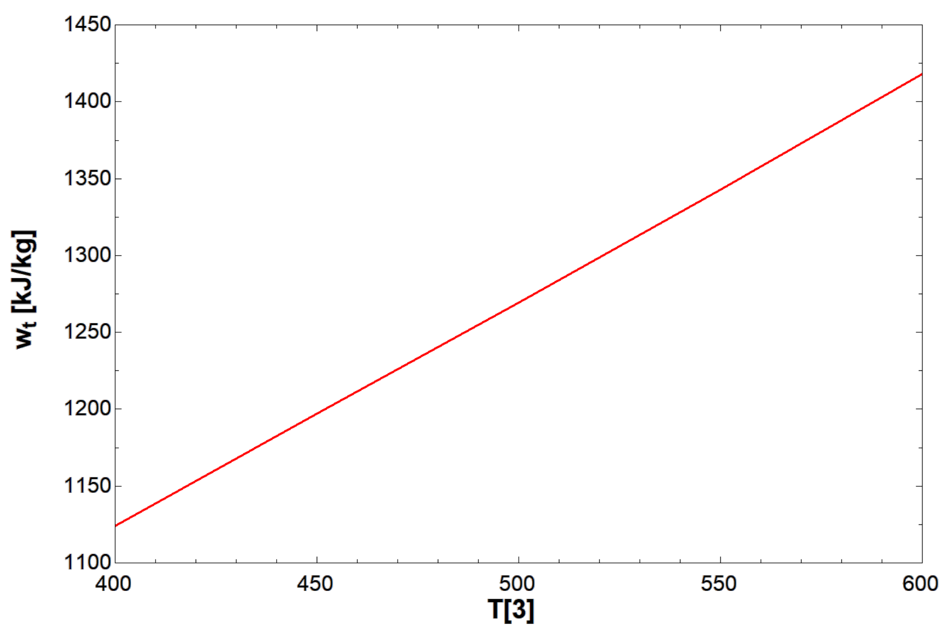

Figure 12: Work done by turbine as affected by a variable temperature leaving the boiler

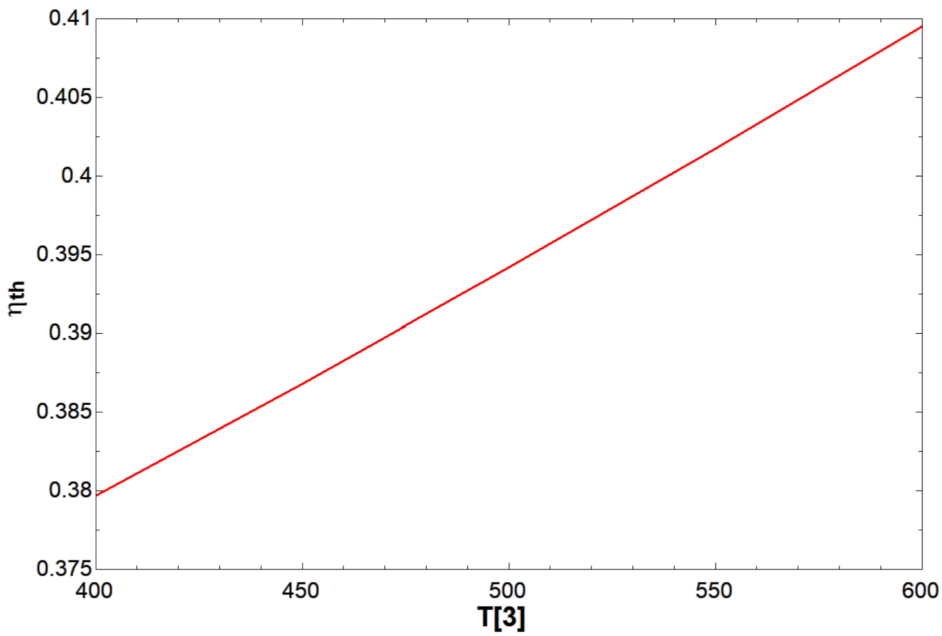

Figure 13: Thermal efficiency as affected by a variable temperature leaving the boiler

5 Discussion

The parametric studies performed on the ideal Rankine cycle certainly revealed some key trends. These trends are evident in Figure 6 and Figure 7, where both turbine work and thermal efficiency increased as pressure exiting the boiler increased. Turbine work eventually reached a constant state once boiler pressure reached 14000kPa, indicting that this pressure property has a non-linear affect on the turbine's mechanical energy output. Meanwhile, the thermal efficiency of the system continuously increased while concaving downward, following a logarithmic path. Thus, this specific parametric study shows an inherent correlation bewteen a boiler pressure increase, with greater turbine work and with improved thermal efficiency.

On the contrary, as evidenced by Figure 10 and Figure 11, turbine work and thermal efficeincy are effectivly inversely proportional to chnages in pressure leaving the condenser. As pressure leaving the condensor is incrementally increased, both the work done by the turbine and the thermal efficiency of the system experience a decline. Thus, this study proves that an increase in pressure at the condenser outlet negatively affects the top two key metrics of a Rankine cycle.

The last parametric study performed was an incremental increase in temperature at the boiler ouput. This study revealed a direct correlation between a temperature increase at the boiler to work output at the turbine and thermal efficiency of the Rankine cycle. Figure 12 and Figure 13 clearly denote that when temperature is increased at the boiler, there is a linear affect/increase on the two notable success mertics, turbine work output and thermal efficiency.

It is worth noting that work done by the turbine and thermal efficiency were chosen as the key metrics in the Rankine cycle, since they both uniquely represent the goals of the cycle: resource conservation and revenue generation through mechanical energy output.

6 Conclusion

The study was successful in its objective of analyzing a Rankine cycle power plant using Engineering Equation Solver. The ideal Rankine cycle used in the study resulted in a thermal efficiency of 39.1% and a net work output of 1230 kJ/kg. The parametric study demonstrated that the pressure and heat of the medium exiting the boiler improved the thermal efficiency and increased the work output of the system. It also demonstrated that an increased pressure of the medium exiting the condenser degraded the thermal efficiency and decreased the work output of the system. The analysis of an ideal Rankine cycle provides interesting insight but may not be completely accurate to real examples like the nuclear reactor at Laguna Verde or the gas power plant in Sayreville. The real models have reduced performance; X and Y could reduce this perfomance loss.

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